Route B: hyperbolic divergence cleaning with a genuine cleaning speed c_h in the GIZMO GRMHD module

Derivation notes for review — revision 2 (code changes implemented, see §7)

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Conventions and notation

Conventions follow Fedrigo & Lupi (2025, A&A 700, A112; arXiv:2506.15775), hereafter FL25: signature $(-,+,+,+)$; $n^\mu = (1/\alpha, -\beta^i/\alpha)$, $n_\mu = (-\alpha, 0, 0, 0)$; $\gamma^{\mu\nu} = g^{\mu\nu} + n^\mu n^\nu$ (so $\gamma^{0\mu} = 0$); $\mathcal{B}^i = \sqrt{\gamma}\, B^i$, $\tilde{v}^i = \alpha v^i - \beta^i$, $D = \sqrt{\gamma}\,\rho W$; K_{ij} is the extrinsic curvature with $K_{ij} = -\alpha\,\Gamma^0_{ij}$ (as in `gr/metric.c`), $K = \gamma^{ij}K_{ij}$, so that $\nabla_\mu n^\mu = -K$; and $a^\mu = n^\nu\nabla_\nu n^\mu = D^\mu \ln\alpha$ is purely spatial ($a^0 = 0$, $a^i = \gamma^{ij}\partial_j \ln\alpha$).

Cleaning scalars. ψ is Dedner's scalar (the one obeying $\partial_t\psi = -c_h^2\nabla\cdot\boldsymbol{B}$ in flat space) and

$$\hat{\psi} \equiv \psi/c_h \tag{1}$$

is the Tricco–Price variable, which is also FL25's $\hat{\psi}$ and the quantity stored by the code (`Phi`). Revision 1 of these notes used the hat for the field multiplying the cleaning tensor and then called ψ/c_h "Ξ"; that was a mislabelling and is corrected here. For $c_h = 1$ the two scalars coincide.

1 The problem with eq. (27) of FL25

FL25 introduce the cleaning field through

$$\nabla_\mu\Big(F^{*\mu\nu} + g^{\mu\nu}\hat{\psi}\Big) = \sigma\, c_h\, \hat{\psi}\, n^\nu. \tag{FL27}$$

Because $g^{\mu\nu}$ is covariantly constant and Lorentz covariant, the $(B^\perp, \hat{\psi})$ subsystem obtained from (FL27) propagates at

$$\lambda_\pm = -\beta^\perp \pm \alpha\sqrt{\gamma^{\perp\perp}}, \tag{2}$$

the coordinate speed of light, for *any* c_h. The cleaning speed survives only through $\hat{\psi} = \psi/c_h$ and the damping term. The Riemann states of FL25 eq. (32) are therefore light-speed states (this is what the live `DIVB_VARYING_CH` branch of `gr/gr_mhd_riemann_solver.h` implements), while the time step and the damping in the code use $c_h = v_{\rm ms}(1+f)/(1+fv_{\rm ms}^2) < 1$. Wherever $v_{\rm ms} \ll 1$ the cleaning wave crosses more than one h per step. To make c_h a genuine propagation speed, the tensor multiplying the scalar must change.

2 Modified covariant equation

Replace $g^{\mu\nu}$ by

$$h^{\mu\nu} \equiv \gamma^{\mu\nu} - c_h^{-2}\, n^\mu n^\nu = g^{\mu\nu} + q\, n^\mu n^\nu, \qquad q \equiv 1 - c_h^{-2}, \tag{3}$$

and write the cleaning system, with Dedner's ψ next to the tensor, as

$$\boxed{\nabla_\mu(F^{*\mu\nu} + h^{\mu\nu}\psi) = \sigma\,\hat{\psi}\, n^\nu = \frac{\sigma}{c_h}\,\psi\, n^\nu} \tag{4}$$

For $c_h = 1$, $h = g$, $\psi = \hat{\psi}$, and (4) is (FL27) including the damping. The right-hand side is normalised so that the damping rate is $\alpha\sigma c_h$ for every c_h (§3.1); FL25's right-hand side, $\sigma c_h \hat{\psi} n^\nu = \sigma \psi n^\nu$, would give a rate $\alpha\sigma c_h^3$ in the h-system.

Components of h:

$$h^{00} = -\frac{c_h^{-2}}{\alpha^2}, \qquad h^{0j} = \frac{c_h^{-2}\beta^j}{\alpha^2}, \qquad h^{ij} = \gamma^{ij} - \frac{c_h^{-2}\beta^i\beta^j}{\alpha^2}. \tag{5}$$

Divergence of h (the new source):

$$\nabla_\mu h^{\mu\nu} = \nabla_\mu (q\, n^\mu n^\nu) = n^\nu \left(n^\mu \partial_\mu q\right) - q\, K\, n^\nu + q\, a^\nu. \tag{6}$$

Flat-space check. With c_h constant, $h = \mathrm{diag}(-c_h^{-2}, 1, 1, 1)$; the $\nu = j$ equations give $\partial_t B^j = -\partial_j \psi$ and the $\nu = 0$ equation gives $\partial_t \psi = -c_h^2\, \partial_i B^i$: the standard Dedner system with characteristic speeds $\pm c_h$. This is what fixes the identification of the field next to $h^{\mu\nu}$ as Dedner's ψ.

3 3+1 reduction

Write $X = \nabla_\mu F^{*\mu 0}$ and $Y^j = \nabla_\mu F^{*\mu j}$. From FL25 eq. (29) one reads off

$$X = -\frac{\partial_i \mathcal{B}^i}{\alpha\sqrt{\gamma}}, \qquad \alpha\sqrt{\gamma}\, Y^j = \partial_t \mathcal{B}^j + \partial_i\left(\tilde{v}^i \mathcal{B}^j - \tilde{v}^j \mathcal{B}^i\right). \tag{7}$$

3.1 $\nu = 0$: the scalar equation

Let $Z \equiv n^\mu \partial_\mu q - qK$. Eq. (4) for $\nu = 0$ reads

$$X - \frac{c_h^{-2}}{\alpha^2}\left(\partial_t - \beta^i \partial_i\right)\psi + \frac{\psi}{\alpha} Z = \frac{\sigma}{c_h}\frac{\psi}{\alpha}. \tag{8}$$

Multiplying by $-\alpha^2 c_h^2$ and using $\alpha c_h^2\, n^\mu \partial_\mu q = 2(\partial_t - \beta^i\partial_i)\ln c_h$:

$$\left(\partial_t - \beta^i\partial_i\right)\psi = -\alpha c_h^2 \frac{\partial_i \mathcal{B}^i}{\sqrt{\gamma}} - \alpha\sigma c_h\, \psi + \psi\Big[\, 2\left(\partial_t - \beta^i\partial_i\right)\ln c_h - \alpha\left(c_h^2 - 1\right) K\,\Big]. \tag{9}$$

In terms of the stored variable $\hat{\psi} = \psi / c_h$ (the change of variable removes one power of the $\ln c_h$ derivative):

$$\left(\partial_t - \beta^i\partial_i\right)\hat{\psi} = -\alpha c_h \frac{\partial_i \mathcal{B}^i}{\sqrt{\gamma}} - \alpha\sigma c_h\, \hat{\psi} - \alpha\left(c_h^2 - 1\right) K\hat{\psi} + \hat{\psi}\left(\partial_t - \beta^i\partial_i\right)\ln c_h. \tag{10}$$

Fluid advection. FL25 do not evolve the covariant scalar equation as such: their eqs. (29)–(30) advect $\hat{\psi}$ with the fluid, $(\partial_t + \tilde{v}^i\partial_i)\hat{\psi}$, in mass-weighted form (GIZMO's "mass-based ϕ flux"). The term $-\sqrt{\gamma}\,\beta^j v^i \partial_i \hat{\psi}$ in their induction equation is the fingerprint of this choice: it appears only if $\partial_t\hat{\psi}$ is eliminated with the fluid-advected equation; the pure covariant equation advects the scalar with $-\beta^i$ only. *To be confirmed by the authors.* The same choice is kept here.

Energy-consistent closure (recommended). The last term of (10) is what the covariant h-system does with a time-dependent c_h: in the cleaning-energy budget (§5) it injects energy where c_h decreases in time. A local c_h (e.g. the maximum magnetosonic speed over the neighbours, re-evaluated every step) is time-dependent by construction and, being a max-filter, discontinuous in time, so the term cannot be neglected and its derivative is not even well defined. Tricco, Price & Bate (2016) obtained their variable-c_h scheme by requiring that the hyperbolic exchange conserve $\int[B^2/2 + \hat{\psi}^2/2]$ for arbitrary $c_h(\boldsymbol{x}, t)$; in the present language that amounts to dropping

the $\ln c_h$-derivative term. Since ψ has no physical content and (4) is only the device that fixes the characteristics, this is the form proposed for implementation:

$$\boxed{\partial_t(D\hat{\psi}) + \partial_i\Big(\tilde{v}^i D\hat{\psi}\Big) = -\alpha c_h D\frac{\partial_i \mathcal{B}^i}{\sqrt{\gamma}} - \alpha\sigma c_h D\hat{\psi} - \alpha\left(c_h^2 - 1\right) K D\hat{\psi}} \tag{11}$$

No derivative of c_h appears, so c_h may be local and re-evaluated every step. For $c_h = 1$, (11) is FL25 eq. (30).

3.2 $\nu = j$: the induction equation

Eq. (4) for $\nu = j$, with $n^j = -\beta^j/\alpha$:

$$Y^j + h^{0j}\partial_t\psi + h^{ij}\partial_i\psi + \psi\, n^j Z + q\,\psi\, a^j = \frac{\sigma}{c_h}\psi\, n^j. \tag{12}$$

Eliminating $\partial_t\psi$ with the fluid-advected form of (9),

$$\partial_t\psi = -\alpha v^i\partial_i\psi + \beta^i\partial_i\psi - \alpha c_h^2\frac{\partial_i\mathcal{B}^i}{\sqrt{\gamma}} - \alpha\sigma c_h\psi + \alpha c_h^2\psi Z, \tag{13}$$

one obtains

$$h^{0j}\partial_t\psi = -\frac{c_h^{-2}\beta^j}{\alpha}v^i\partial_i\psi + \frac{c_h^{-2}\beta^i\beta^j}{\alpha^2}\partial_i\psi - \frac{\beta^j\partial_i\mathcal{B}^i}{\alpha\sqrt{\gamma}} - \frac{\beta^j}{\alpha}\frac{\sigma}{c_h}\psi + \frac{\beta^j}{\alpha}\psi Z, \tag{14}$$

$$h^{ij}\partial_i\psi = \gamma^{ij}\partial_i\psi - \frac{c_h^{-2}\beta^i\beta^j}{\alpha^2}\partial_i\psi. \tag{15}$$

The $\beta^i\beta^j$ terms cancel; $\psi n^j Z = -(\beta^j/\alpha)\psi Z$ cancels the Z term; and the damping term cancels against the right-hand side, exactly as in the $c_h = 1$ case. Multiplying what is left by $\alpha\sqrt{\gamma}$ and writing $\psi = c_h\hat{\psi}$:

$$\boxed{\partial_t\mathcal{B}^j + \partial_i\big(\tilde{v}^i\mathcal{B}^j - \tilde{v}^j\mathcal{B}^i\big) = -\sqrt{\gamma}\left(c_h^{-2}\beta^j v^i + \alpha\gamma^{ij}\right)\partial_i(c_h\hat{\psi}) + \beta^j\partial_i\mathcal{B}^i - \sqrt{\gamma}\left(1 - c_h^{-2}\right)c_h\hat{\psi}\,\gamma^{jk}\partial_k\alpha} \tag{16}$$

For $c_h = 1$, (16) is FL25 eq. (29). The only changes are the factor c_h^{-2} on the $\beta^j v^i$ term and the lapse-gradient source. Two remarks: (i) $\partial_i(c_h\hat{\psi})$ needs no explicit ∇c_h: in the code the gradient is assembled from face values of $\psi = c_h\hat{\psi}$, exactly as the ϕ gradient is assembled today. (ii) If one re-did the elimination of $\partial_t\psi$ with the closure (11) instead of (9), a term $\propto \beta^j\hat{\psi}\,(\partial_t - \beta^i\partial_i)\ln c_h$ would be generated; it is dropped together with the other c_h-derivative terms, consistently with the closure (it is not a total divergence in the energy budget).

4 Riemann states for the $(\mathcal{B}^\perp, \psi)$ subsystem

From (11)–(16), the subsystem along the face normal $\perp$ has fluxes $F_B = \alpha\sqrt{\gamma}\gamma^{\perp\perp}\psi$ and $F_\psi = \alpha c_h^2\mathcal{B}^\perp/\sqrt{\gamma}$, hence characteristic speeds $\lambda = \pm\alpha c_h\sqrt{\gamma^{\perp\perp}}$ (plus the advection carried by the full system). Rusanov fluxes with $|\lambda| = \alpha c_h\sqrt{\gamma^{\perp\perp}}$ give, with $\psi_{L,R} = c_h^{L,R}\hat{\psi}_{L,R}$ and c_h at the face the larger of the two particle values (as in FL25),

$$\bar{\mathcal{B}}^\perp = \tfrac{1}{2}\left(\mathcal{B}_L + \mathcal{B}_R\right) - \tfrac{1}{2}\sqrt{\gamma\gamma^{\perp\perp}}\,\frac{\psi_R - \psi_L}{c_h}, \tag{17}$$

$$\bar{\psi} = \tfrac{1}{2}\left(\psi_L + \psi_R\right) - \tfrac{1}{2}\,c_h\left(\gamma\gamma^{\perp\perp}\right)^{-1/2}\left(\mathcal{B}_R - \mathcal{B}_L\right), \tag{18}$$

which reduce to FL25 eq. (32) for $c_h = 1$. Note the powers: $1/c_h$ in (17) and c_h in (18), *not* c_h^2 as in the dead `#ifndef DIVB_VARYING_CH` branch of the solvers.

In terms of the primitive $B = \mathcal{B}/\sqrt{\gamma}$ used inside the solver, $\bar{B} = \frac{1}{2}(B_L + B_R) + \frac{1}{2}\sqrt{\gamma^{\perp\perp}}\,(\psi_L - \psi_R)/c_h$, i.e. the present `corr_p` with $\hat{\psi} \to c_h\hat{\psi}$ on each side and divided by the face c_h; the present `phi_normal_db` multiplied by the face c_h; and `phi_normal_mean` built from $\psi_{L,R} = c_h^{L,R}\hat{\psi}_{L,R}$.

5 Energy and momentum

Unchanged in form: with Ξ^k the full right-hand side of (16) minus the ideal part, the sources of FL25 eq. (34) (the chain rule $\partial\tau/\partial B^k\,\Xi^k$, $\partial S_j/\partial B^k\,\Xi^k$) keep the fluid primitives unaffected by the cleaning. τ is not conserved by the cleaning, now as before.

Cleaning energy, flat space, hyperbolic part only, $E_\psi = \int \left[B^2/2 + \hat{\psi}^2/2\right] = \int \left[B^2/2 + \psi^2/(2c_h^2)\right]$: with the covariant equation (10),

$$\frac{dE_\psi}{dt} = -\oint \psi\, \boldsymbol{B}\cdot d\boldsymbol{A} + \int \hat{\psi}^2\, \partial_t \ln c_h \;-\; \int \alpha\sigma c_h \hat{\psi}^2 \quad (\le 0,\ \text{damping}), \tag{19}$$

so a c_h that varies in time injects energy where it decreases; with the closure (11) the second term is absent and the exchange is conservative for arbitrary $c_h(\boldsymbol{x}, t)$, the damping monotonically dissipative. In the discrete scheme the exchange is only approximately antisymmetric when $c_h^L \neq c_h^R$, because the face $\bar{\psi}$ uses a pair value; this is the same approximation the Newtonian GIZMO cleaning makes. With $\hat{\psi}$ as the stored variable, a jump of c_h between two steps leaves $\hat{\psi}^2/2$ unchanged; with ψ stored it would not.

6 Choice of c_h

With the closure (11) no smoothness of c_h is required, so FL25's local choice (maximum over the interacting pair of $v_{\rm ms}(1+f)/(1+fv_{\rm ms}^2)$, re-evaluated every step) is admissible and keeps the time-step hierarchy. Two additions are suggested: (i) a floor $c_h \to 1$ near the horizon, since the K term of (11), $-\alpha(c_h^2-1)K\hat{\psi}$, is a growth term for $c_h < 1$ where $K > 0$ (ingoing Kerr–Schild slicing), and the lapse-gradient term of (16) is largest there; (ii) the single function `gr_cleaning_speed()` used by the solver (face value), the scalar source, the damping and the Courant condition, so that all four see the same number.

7 Mapping to the code (implemented 12 Oct 2026, for review)

Item	File	Change
single definition of c_h	`gr_cleaning_speed()` in `gr/gr.h` (implemented)	replaces the `VEL_CLEANING` boost in `hydro/hydro_evaluate.h:273-277`, `c_h2` in `hydro/hydro_toplevel.c:803-807`, damping in `predict.c:550-557`
face states (17)–(18)	`gr/gr_mhd_riemann_solver.h`: HLL :118-131, GForce :433-446, HLLC :756-763	$\psi_{L,R} = c_h^{L,R}$ `phi`$_{L,R}$; `corr_p /= c_h`; `phi_normal_db *= c_h`; delete the `ch2` logic; $c_h^{L,R}$ passed via `Riemann_vec`
$\hat{\psi}$ source (11)	`hydro/hydro_toplevel.c:892-900`	`tmp_ded*tmp_ded` $\to$ `c_h` (linear in c_h)
K term (11)	`hydro/hydro_toplevel.c`, cleaning block	`DtPhi -= alpha (c_h^2-1) K Phi`; K from `get_metric_derivatives` (`gr/metric.c:365`)
$\beta^j v^i$ term (16)	`hydro/hydro_toplevel.c:871-876`	multiply by c_h^{-2}
lapse-gradient term (16)	`hydro/hydro_toplevel.c`, cleaning block	`DtB[j] -= gdet (1-c_h^-2) c_h Phi gamma^jk dalpha_k`, fed through `corr_tau/corr_S`
damping	`predict.c:556`	rate $\alpha\sigma c_h$ with $\sigma = K/\Delta x$ (restores the $1/h$ of the dead branch)
time step	`hydro/hydro_evaluate.h`, `timestep.c`	`vsig = max(vsig_fluid, 2 c_h)`
retire	`allvars.h:946`, `gr/gr.h:24-26`	`DIVB_VARYING_CH`, `VEL_CLEANING_MAX`; keep f

Regression requirement: with $c_h = 1$ everywhere all new terms vanish and the code must reproduce the current results to round-off.

Points to confirm

1. The $\beta^j v^i \partial_i \hat{\psi}$ term in FL25 eq. (29) arises only if $\partial_t \hat{\psi}$ is eliminated using the fluid-advected scalar equation (eq. 30); the pure covariant eq. (27) advects the scalar with $-\beta^i$ only. That choice is kept here.

2. The damping normalisation: $\sigma\hat{\psi}n^\nu$ on the right-hand side of (4) keeps the rate at $\alpha\sigma c_h$; FL25's $\sigma c_h \hat{\psi} n^\nu$ would give $\alpha\sigma c_h^3$ in the h-system.

3. The sign convention $K_{ij} = -\alpha\Gamma^0_{ij}$ in `gr/metric.c:365` is the one for which $\nabla_\mu n^\mu = -K$; with ingoing Kerr–Schild slicing $K > 0$ near the horizon, which makes the K term a growth term for $c_h < 1$, hence the floor $c_h \to 1$ there.

4. The closure: all terms containing derivatives of c_h are dropped, in both the scalar and the induction equation, in exchange for exact (continuum) energy consistency of the hyperbolic exchange for any $c_h(\boldsymbol{x}, t)$. For constant c_h the closure and the covariant system coincide.